```python
# LEVEL 1 - TASK 1.1

import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python')

print("\n========= FIRST 5 ROWS =========\n")
print(df.head())
print("\n========= LAST 5 ROWS =========\n")
print(df.tail())

print("\n========= DATASET SHAPE =========\n")
print("Total Records :", df.shape[0])
print("Total Columns :", df.shape[1])

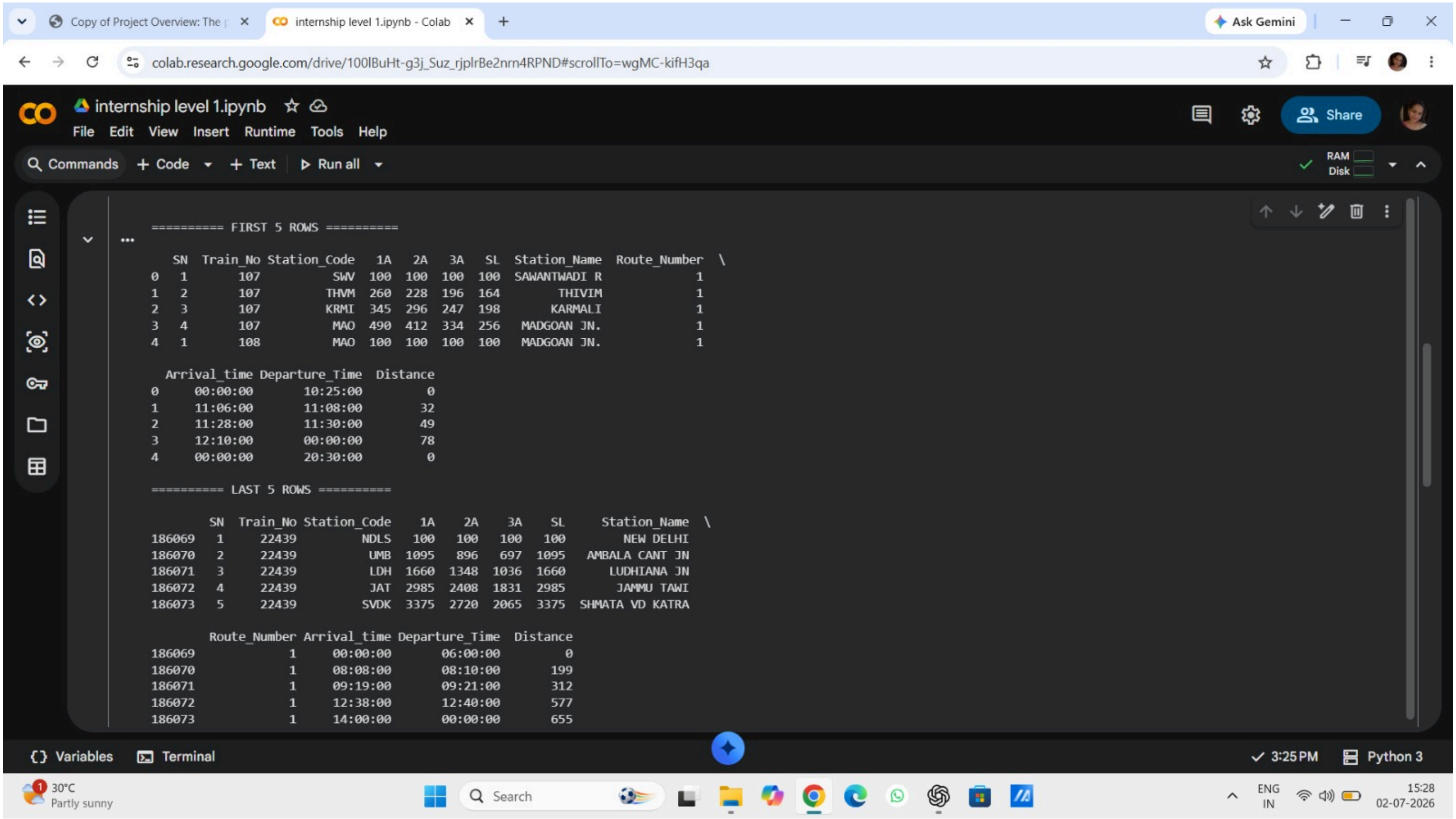
print("\n========= COLUMN NAMES =========\n")
print(df.columns)

print("\n========= DATA TYPES =========\n")
print(df.dtypes)

print("\n========= DATASET INFORMATION =========\n")
df.info()

print("\n========= DESCRIPTIVE STATISTICS =========\n")
print(df.describe())

print("\nTask 1.1 Completed Successfully.")
```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internship level 1.ipynb
File Edit View Insert Runtime Tools Help
Commands + Code + Text Run all
RAM
Disk
Share
========== FIRST 5 ROWS ==========
SN Train_No Station_Code 1A 2A 3A SL Station_Name Route_Number \
0 1 107 SWV 100 100 100 100 SAWANTWADI R 1
1 2 107 THVM 260 228 196 164 THIVIM 1
2 3 107 KRMI 345 296 247 198 KARMALI 1
3 4 107 MAO 490 412 334 256 MADGOAN JN. 1
4 1 108 MAO 100 100 100 100 MADGOAN JN. 1
Arrival_time Departure_Time Distance
0 00:00:00 10:25:00 0
1 11:06:00 11:08:00 32
2 11:28:00 11:30:00 49
3 12:10:00 00:00:00 78
4 00:00:00 20:30:00 0
========== LAST 5 ROWS ==========
SN Train_No Station_Code 1A 2A 3A SL Station_Name \
186069 1 22439 NDLS 100 100 100 100 NEW DELHI
186070 2 22439 UMB 1095 896 697 1095 AMBALA CANT JN
186071 3 22439 LDH 1660 1348 1036 1660 LUDHIANA JN
186072 4 22439 JAT 2985 2408 1831 2985 JAMMU TAWI
186073 5 22439 SVDK 3375 2720 2065 3375 SHMATA VD KATRA
Route_Number Arrival_time Departure_Time Distance
186069 1 00:00:00 06:00:00 0
186070 1 08:08:00 08:10:00 199
186071 1 09:19:00 09:21:00 312
186072 1 12:38:00 12:40:00 577
186073 1 14:00:00 00:00:00 655
Variables
Terminal
3:25 PM
Python 3
30°C
Partly sunny
Search
ENG
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02-07-2026

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========= DATASET SHAPE =========

Total Records : 186074
Total Columns : 12

========= COLUMN NAMES =========

Index(['SN', 'Train_No', 'Station_Code', '1A', '2A', '3A', 'SL',
       'Station_Name', 'Route_Number', 'Arrival_time', 'Departure_Time',
       'Distance'],
      dtype='object')

========= DATA TYPES =========

SN                 int64
Train_No           int64
Station_Code      object
1A                 int64
2A                 int64
3A                 int64
SL                 int64
Station_Name      object
Route_Number       int64
Arrival_time      object
Departure_Time    object
Distance           int64
dtype: object

========= DATASET INFORMATION =========

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 186074 entries, 0 to 186073
Data columns (total 12 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
```

Variables | Terminal | 3:25 PM | Python 3

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Commands | + Code | + Text | Run all | RAM Disk

```
========= DATASET INFORMATION =========

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 186074 entries, 0 to 186073
Data columns (total 12 columns):
 #   Column          Non-Null Count   Dtype
---  ------          --------------   -----
 0   SN              186074 non-null  int64
 1   Train_No        186074 non-null  int64
 2   Station_Code    186074 non-null  object
 3   1A              186074 non-null  int64
 4   2A              186074 non-null  int64
 5   3A              186074 non-null  int64
 6   SL              186074 non-null  int64
 7   Station_Name    186074 non-null  object
 8   Route_Number    186074 non-null  int64
 9   Arrival_time    186074 non-null  object
 10  Departure_Time  186074 non-null  object
 11  Distance        186074 non-null  int64
dtypes: int64(8), object(4)
memory usage: 17.0+ MB

========= DESCRIPTIVE STATISTICS =========

                  SN       Train_No             1A             2A  \
count  186074.000000  186074.000000  186074.000000  186074.000000
mean       13.914695   42157.747455    1506.769189    1225.415351
std        12.779368   25090.080221    2418.719821    1934.975856
min         1.000000     107.000000     100.000000     100.000000
25%         5.000000   17225.000000     215.000000     192.000000
50%        11.000000   40050.000000     465.000000     392.000000
75%        18.000000   57550.000000    1555.000000    1264.000000
max       118.000000   99908.000000   21400.000000   17140.000000

                  3A             SL  Route_Number       Distance
```

Variables | Terminal | 3:25 PM | Python 3

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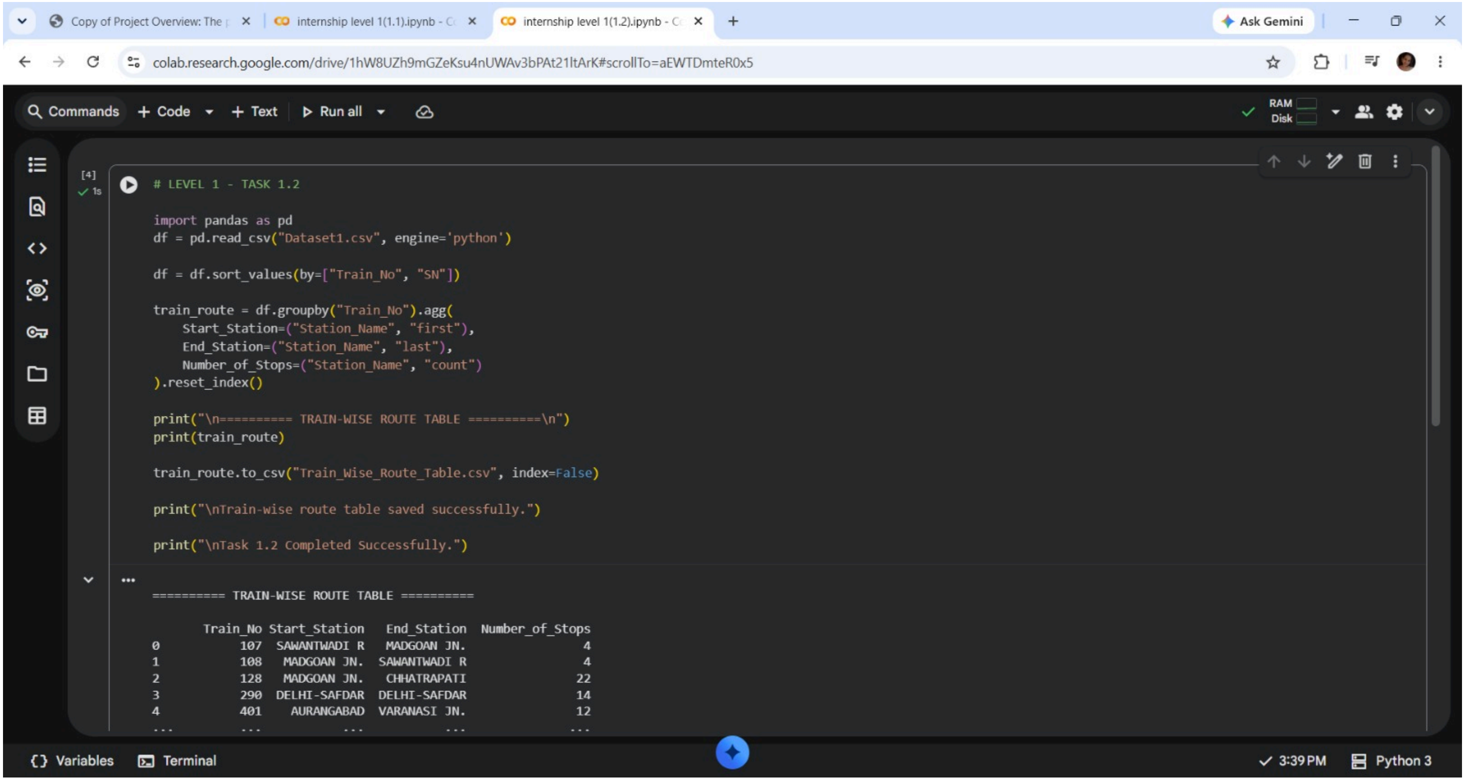
```
========== DESCRIPTIVE STATISTICS ==========

                 SN       Train_No             1A             2A  \
count  186074.000000  186074.000000  186074.000000  186074.000000
mean       13.914695   42157.747455    1506.769189    1225.415351
std        12.779368   25090.080221    2418.719821    1934.975856
min         1.000000     107.000000     100.000000     100.000000
25%         5.000000   17225.000000     215.000000     192.000000
50%        11.000000   40050.000000     465.000000     392.000000
75%        18.000000   57550.000000    1555.000000    1264.000000
max       118.000000   99908.000000   21400.000000   17140.000000

                 3A             SL  Route_Number       Distance
count  186074.000000  186074.000000      186074.0  186074.000000
mean      944.061513     662.735777           1.0     281.353838
std      1451.231892     967.524012           0.0     483.743964
min       100.000000     100.000000           1.0       0.000000
25%       169.000000     146.000000           1.0      23.000000
50%       319.000000     246.000000           1.0      73.000000
75%       973.000000     682.000000           1.0     291.000000
max     12880.000000    8620.000000           1.0    4260.000000

Task 1.1 Completed Successfully.
```

Variables Terminal ✓ 3:25 PM Python 3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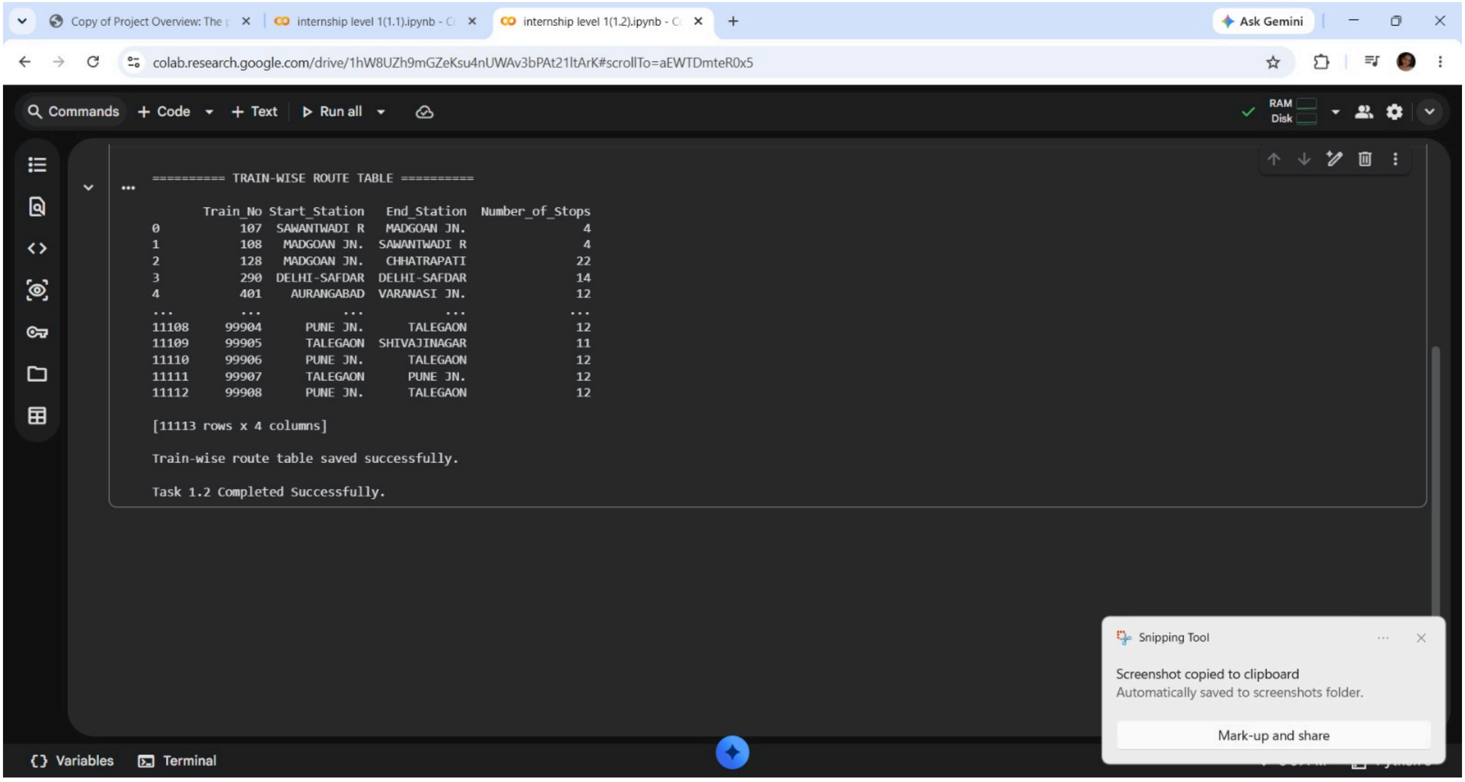
Commands
Code
Text
Run all
RAM
Disk
[4]
1s
# LEVEL 1 - TASK 1.2
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python')
df = df.sort_values(by=["Train_No", "SN"])
train_route = df.groupby("Train_No").agg(
    Start_Station=("Station_Name", "first"),
    End_Station=("Station_Name", "last"),
    Number_of_Stops=("Station_Name", "count")
).reset_index()
print("\n========= TRAIN-WISE ROUTE TABLE =========\n")
print(train_route)
train_route.to_csv("Train_Wise_Route_Table.csv", index=False)
print("\nTrain-wise route table saved successfully.")
print("\nTask 1.2 Completed Successfully.")
========= TRAIN-WISE ROUTE TABLE =========
   Train_No Start_Station   End_Station Number_of_Stops
0   107   SAWANTWADI R   MADGOAN JN.   4
1   108   MADGOAN JN. SAWANTWADI R   4
2   128   MADGOAN JN.   CHHATRAPATI   22
3   290   DELHI-SAFDAR DELHI-SAFDAR   14
4   401   AURANGABAD VARANASI JN.   12
...   .
Variables
Terminal
3:39 PM
Python 3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internship level 1(1.2).ipynb - C
Ask Gemini
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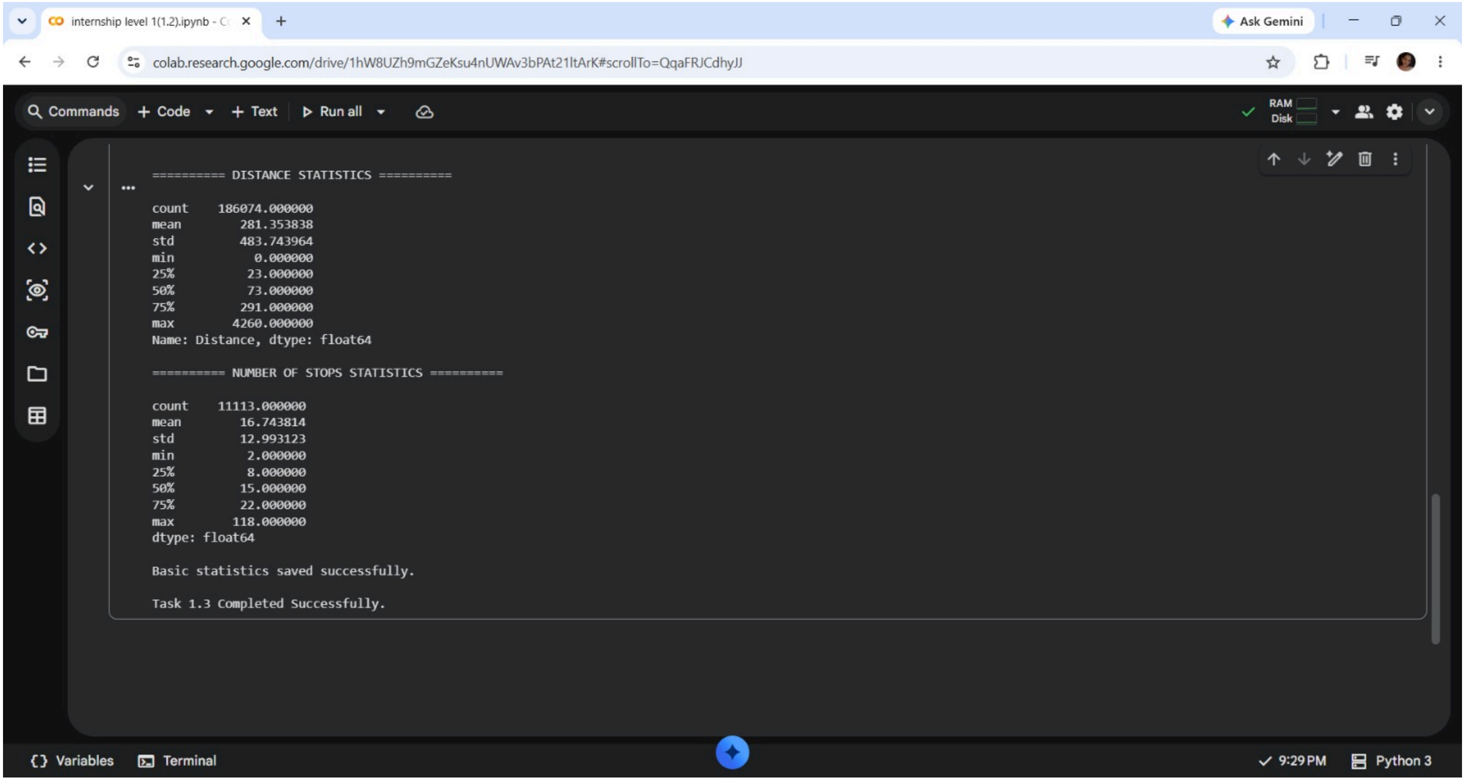
Commands
Code
Text
Run all
RAM
Disk
========== TRAIN-WISE ROUTE TABLE ==========
Train_No Start_Station End_Station Number_of_Stops
0 107 SAWANTWADI R MADGOAN JN. 4
1 108 MADGOAN JN. SAWANTWADI R 4
2 128 MADGOAN JN. CHHATRAPATI 22
3 290 DELHI-SAFDAR DELHI-SAFDAR 14
4 401 AURANGABAD VARANASI JN. 12
... ...
11108 99904 PUNE JN. TALEGAON 12
11109 99905 TALEGAON SHIVAJINAGAR 11
11110 99906 PUNE JN. TALEGAON 12
11111 99907 TALEGAON PUNE JN. 12
11112 99908 PUNE JN. TALEGAON 12
[11113 rows x 4 columns]
Train-wise route table saved successfully.
Task 1.2 Completed Successfully.
Snipping Tool
Screenshot copied to clipboard
Automatically saved to screenshots folder.
Mark-up and share
Variables
Terminal

```python
# LEVEL 1 - TASK 1.3
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python', on_bad_lines='skip')

print("\n========= DISTANCE STATISTICS =========\n")
print(df["Distance"].describe())

stops = df.groupby("Train_No").size()

print("\n========= NUMBER OF STOPS STATISTICS =========\n")
print(stops.describe())


distance_stats = df["Distance"].describe()
stop_stats = stops.describe()


statistics = pd.DataFrame({
    "Distance": distance_stats,
    "Number_of_Stops": stop_stats
})

statistics.to_csv("Basic_Statistics.csv")


print("\nBasic statistics saved successfully.")


print("\nTask 1.3 Completed Successfully.")
```

```
========= DISTANCE STATISTICS =========
```




```
========== DISTANCE STATISTICS ==========

count    186074.000000
mean        281.353838
std         483.743964
min           0.000000
25%          23.000000
50%          73.000000
75%         291.000000
max        4260.000000
Name: Distance, dtype: float64

========== NUMBER OF STOPS STATISTICS ==========

count    11113.000000
mean        16.743814
std         12.993123
min          2.000000
25%          8.000000
50%         15.000000
75%         22.000000
max        118.000000
dtype: float64

Basic statistics saved successfully.

Task 1.3 Completed Successfully.
```

```python
# LEVEL 1 - TASK 1.4
import pandas as pd
df = pd.read_csv("Dataset1.csv", engine='python')

missing_values = df.isnull().sum()

duplicate_records = df.duplicated().sum()

empty_station_names = (df["Station_Name"].astype(str).str.strip() == "").sum()

negative_distance = (df["Distance"] < 0).sum()

summary = pd.DataFrame({
    "Check": [
        "Missing Values",
        "Duplicate Records",
        "Empty Station Names",
        "Negative Distance Values"
    ],
    "Count": [
        missing_values.sum(),
        duplicate_records,
        empty_station_names,
        negative_distance
    ]
})

print("\n========= MISSING VALUES =========\n")
print(missing_values)

print("\n========= DATA QUALITY SUMMARY =========\n")
print(summary)
```

```python
summary.to_csv("Data_Quality_Summary.csv", index=False)

print("\nData quality summary saved successfully.")

print("\nTask 1.4 Completed Successfully.")
```

```
========== MISSING VALUES ==========

SN                0
Train_No          0
Station_Code      0
1A                0
2A                0
3A                0
SL                0
Station_Name      0
Route_Number      0
Arrival_time      0
Departure_Time    0
Distance          0
dtype: int64

========== DATA QUALITY SUMMARY ==========

                      Check  Count
0            Missing Values      0
1         Duplicate Records      0
2       Empty Station Names      0
3  Negative Distance Values      0

Data quality summary saved successfully.

Task 1.4 Completed Successfully.
```